Correlation and Causation for Attention in a Sepsis-Forecasting Transformer

Eight-point methodology (slides 7 and 8), with journal cross-check and a full explanation of DAS

How to read this. Slide 7 has four correlational points (P1–P4) and slide 8 has four causal points (P5–P8). Each point below gives: what it is, why (evidence), the exact procedure, and how it can fail. Every claim carries a tag: **[C] cited** — a finding from a journal listed in the references; **[D] derived** — our own algebra, not a citation; **[A] analogy / assumption** — a project decision that the literature does not establish. Notation: t = readout (query) hour, k = the hour we intervene on, j = any visible key hour (j ≤ t), z = pre-softmax logit, α = attention weight.

Inside. Part A: P1–P4 (correlation). Part B: P5–P8 (causation), including the research-backed high-level model, **what γ is** (B0a), the patch sites and a full explanation of DAS. **Part C: how the design meets ABC's contested evidence. Part D: the sigmoid-arm analysis.** Then slide edits, limits and references.

What the literature cross-check changed (read this first)

- **Sign of the selectivity test.** With the DV = Δ log α (or Δz), ABC-style selective suppression predicts a slope > 0 on priority (low-priority hours fall below the common shift, high-priority hours above it). Earlier notes said "slope < 0"; that is correct only if the DV is *suppression* = −Δ. Pre-register one convention and use it everywhere (spreadsheet row 1 included).
- **The earlier high-level model was the null model.** Budget + Share with a fixed exponent γ means do(Priority$_k$) rescales every other hour by the same factor ($r_j \equiv 0$). A high IIA on it would confirm plain normalization, not ABC. ABC adds *arousal-amplified* competition, so the graph needs a gain node; we now test three rival models (M0, M1, M2) instead of one.
- **γ is not identified on its own.** $\log \alpha_j = \gamma \cdot \log P_j + \kappa$ only identifies the product γ·f(severity). Test *changes* in sharpness under intervention, not "γ > 1" as an absolute level.
- **Slide 7, point 3 is out of date.** It says mixed-effects with a patient random effect; the design that is computationally feasible is two-stage (per-stay fit, then random-effects meta-analysis). Suggested slide text is on the last pages.
- **ABC itself is empirically contested** (Section B0). The project tests a hypothesis; a null result is a valid, reportable outcome.

1. Why two stages

Jain & Wallace (2019) showed that attention weights often fail to track other importance measures and that very different weights can give the same prediction; Wiegreffe & Pinter (2019) argued their tests do not settle the question. **[C]** Both sides argue from correlational or post-hoc evidence. For a softmax model there is an additional, specific reason correlation cannot decide: the weights in a row sum to 1, so a rise on one hour *forces* a fall elsewhere whether or not the network learned anything selective. **[D]**

Stage 1 (correlation) therefore does three jobs only: it checks the model is non-degenerate, it establishes a noise floor with controls, and it picks which heads and layers are worth intervening on. Stage 2 (causation) is the test, because only an intervention can separate learned competition from normalization arithmetic.

Order of the eight points. Stage 1 screens and sets the noise floor (P1–P4); Stage 2 intervenes (P5–P8).

Stage	Manipulates	Claim it can support	Over-reading to avoid
Correlation (P1–P4)	Nothing; observes natural attention	Attention is severity-sensitive after controlling position, recency, density; heads worth testing	Treating a positive coefficient as support for ABC; any monotone model produces it
Causation (P5–P8)	Inputs, activations, or logits	Changing X changes attention allocation in the shape a specific high-level model predicts	Treating localization (chosen on LOC) as confirmation; letting a learned rotation do the computation itself

Part A — Correlational design (slide 7, points P1–P4)

Pipeline. Capture z and α → per-stay regression with controls (P1–P3) → run the identical pipeline on control networks and control inputs (P4) to give the noise floor. Δz belongs to the causal stage.

P1 · Attention as the dependent variable

What and why

Under a causal mask, row *t* of a softmax head has only *t* entries, so its mean weight is 1/t by construction; severity and onset also correlate with position in a stay, so raw weights confound position with severity. **[D]** A row of weights is compositional data (positive, fixed sum). Aitchison (1982) established log-ratio transforms for exactly this situation. **[C]** For softmax, $\alpha_j = e^{z_j}/Z$, therefore $\mathrm{clr}(\alpha)_j = \log \alpha_j - \mathrm{mean}_i \log \alpha_i = z_j - \mathrm{mean}_i z_i$: the normalizer Z cancels exactly. **[D]**

Heads differ: Voita et al. (2019) and Michel et al. (2019) found that a minority of heads do most of the work and many can be pruned, so we analyse **per head**, never averaged, and choose heads on a discovery split. **[C]** Jain & Wallace (2019) is the reason we do not stop at weights.

Procedure

- Capture z (pre-normalizer) and α for every layer and head; keep only $j \le t$ (drop $-\infty$ entries and padded positions).
- **Screening DV (softmax arm):** clr(α). **Sigmoid arm:** $\log a_{tj}$, uncentered, because there is no Z to cancel; keep a query-position covariate, since the $-\log T$ bias is a deliberate t-dependent term. **[D]**
- **Causal DV (both arms):** Δz = z(source) − z(base), captured before either normalizer runs, so it is the same quantity in both arms (used in P5–P7).
- Covariates per row: $\mathrm{severity}_j$ (SOFA_total), $\log(t-j+1)$, pool size $\min(t+1, W)$, measurement density, deviation magnitude, and ICULOS_t only when stays are truncated or windowed ($\mathrm{ICULOS}_j = \mathrm{ICULOS}_t - (t-j)$, so including all three time terms is perfectly collinear).

How it can fail

Recency is closer to a power law than a line (use $\log(t-j+1)$ or a spline); severity and deviation are nearly collinear (check VIF, residualize by a pre-registered rule); in layer 1 keys depend only on the hour's own embedding, so any intervention-driven structure on other hours cannot appear before layer 2.

P2 · Norm-weighted attention

What and why

A key can have a large weight and still contribute little if its value vector is small. Kobayashi et al. (2020) showed that weight alone can misrepresent a token's contribution and proposed weight × ‖value vector‖. **[C]** Kobayashi et al. (2021) extended this to include the residual connection and LayerNorm, which matters in a Pre-LN model because the residual stream dominates. **[C]**

Procedure

- For head h, contribution $n_{tj}^{(h)} = \alpha_{tj}^{(h)} \cdot \| v_j^{(h)} W_O^{(h)} \|_2$, where $W_O^{(h)}$ is the head's slice of the output projection. The value tensor is already captured by the attention module.
- Secondary DV. If clr(α) and n_{tj} tell different stories, report both rather than choosing one; that disagreement is itself the Jain & Wallace warning showing up in your model.
- Optional Pre-LN variant: report n_{tj} relative to the residual-stream norm at hour t (Kobayashi 2021 logic).

P3 · Two-stage regression (replaces "mixed-effects" on slide 7)

What and why

A single multilevel model needs every (t, j) pair: about 5×10^3 rows per stay per head-layer at T = 100, over 48 head-layer cells and thousands of stays — 10^8–10^9 rows. Instead, fit a regression **per stay**, then meta-analyse the per-stay coefficients with a random-effects model (DerSimonian & Laird, 1986). **[C]** Burke, Ensor & Riley (2017) compare one-stage and two-stage individual-participant-data analyses and explain when they differ; with thousands of units they typically agree. **[C]** Gelman & Hill (2007) give the multilevel logic for nested data, and lme4 (Bates et al., 2015) can be run as a one-stage sensitivity check on a small subsample (for example 300 stays). **[C]**

Specification

```
per stay s, head h, layer l, rows (t, j<=t):
  DV_tj = b0 + b1*severity_j + b2*log(t-j+1) + b3*pool_t + b4*density_j + b5*deviation_j [+ b6*ICULOS_t]
pool_t = t+1 (full history)  or  min(t+1, W) (windowed config)
stage 2: random-effects meta-analysis of b1_hat_s across stays -> pooled b1, 95% CI, tau^2
```

- **Unit of inference = the stay** (subject-level if a subject has several stays: average or cluster).
- **Headline = standardized b1 with CI**, not p; at this n, $p < 0.05$ is trivially met. Pre-register a smallest effect size of interest and test nulls with equivalence tests (TOST; Lakens, 2017). **[C]**
- **Head selection:** rank heads on DISCOVERY, confirm on a disjoint SCREEN split; carry the top-k forward (pre-register k).

P4 · Controls before trusting any effect

What and why

A correlation only means something relative to what the same pipeline gives when the claimed ingredient is absent. Adebayo et al. (2018) introduced parameter-randomization and data-randomization sanity checks for explanation methods: if the output barely changes with random weights or random labels, it was not explaining the model. **[C]** Hewitt & Liang (2019) use control tasks to separate what a probe finds from what the probe could fit anyway. **[C]** Theiler et al. (1992) introduced surrogate-data testing for time series; a circular shift keeps autocorrelation and the marginal distribution while destroying alignment. **[C]** Haviv et al. (2022) found that causal language models can recover positional information without explicit positional encodings, so position cannot be removed by omission — it has to be controlled. **[C]**

Control	What it breaks	Reading
Untrained network (same architecture, PE, mask)	All learned structure	If it shows the severity effect, the effect is architectural or positional
Shuffled-label model	Clinical signal; keeps feature structure	Separates "learned physiology" from "learned input statistics"

Control	What it breaks	Reading
Circular-shift severity within stay	Alignment; keeps autocorrelation	The correct null for smooth signals (plain shuffling is too weak)
Donor severity from another stay	Link to this stay	Same marginals, wrong patient
Random-projection severity	Meaning of the construct	Matched marginals, arbitrary content

Decision rule (pre-register): a coefficient counts only if it exceeds the control band (for example the 95th percentile of control β) *and* the smallest effect of interest. **Gate first:** held-out AUPRC must be well above the base rate before any interpretability run; if it fails, fix the model, do not analyse it.

Part B — Causal design (slide 8, points P5–P8)

B0 · Why the high-level model must be research-backed, and what ABC does and does not give us

In causal abstraction (Geiger et al., 2021; 2025; formal basis in Beckers & Halpern, 2019), a high-level causal model is a hypothesis about the algorithm the network implements; the claim is supported to the degree that interventions on aligned variables have matching effects. **[C]** The high-level model is chosen by the researcher, so its credibility depends on being grounded independently of the network. Sutter et al. (2025) show why this matters: with unconstrained (nonlinear) alignment maps, any network can be aligned with any algorithm, and even randomly initialized networks reach high interchange accuracy. **[C]** Hence: ground M_high in established theory, restrict the alignment map, and always run random-network controls.

What ABC contributes. Mather & Sutherland (2011) propose that arousal amplifies competition: high-priority stimuli win more and low-priority stimuli lose more ("winner-take-more, loser-take-less"), building on biased-competition and saliency models (Bundesen, 1990; Desimone & Duncan, 1995; Itti & Koch, 2000). **[C]** Mather et al. (2016) give a neural account (norepinephrine amplifying selectivity). **[C]** Sutherland & Mather (2012) and Lee, Itti & Mather (2012) report supporting behavioural evidence. **[C]** Reynolds & Heeger (2009) formalize attention as gain on the stimulus drive followed by divisive normalization by a suppressive pool; the normalization form goes back to Heeger (1992) and is reviewed as a canonical computation by Carandini & Heeger (2012). **[C]** Eldar, Cohen & Niv (2013) link global neural gain to focus versus breadth of attention. **[C]**

Honest limits. Mapping hour-level clinical severity onto "priority" and clinical acuity onto "arousal" is a project analogy. **[A]** ABC's selective-suppression prediction is also contested: Keitel et al. (2013) found that competition suppressed attended and unattended stimuli similarly while attention acted as an independent gain, and a Leiden University group reports no arousal-biased competition in focused visuospatial attention (partial citation, see references). **[C]** The second report also raises a measurement point close to our own: interaction patterns can be hard to interpret without the right null. Our proportional-renormalization null plays that role.

Research-backed high-level model. Solid structure follows Reynolds & Heeger (2009) and the ABC literature; nodes tagged A are project analogies; equivalences tagged D are our algebra. The table below defines each node.

Node	Formula / definition	Literature basis	Tag
Severity_j	SOFA_total at hour j	Vincent et al. 1998 (SOFA vs mortality dose-response); Singer et al. 2016 (Sepsis-3)	C
Priority_j	$\log P_j = g(\text{Severity}_j)$, monotone	ABC priority = bottom-up salience (Itti & Koch 2000) + top-down relevance (Desimone & Duncan 1995; Miller & Cohen 2001); Mather & Sutherland 2011	C concept; A clinical mapping
Budget_t	$\Sigma_{i \le t} P_i^{\gamma}$	Limited capacity (Desimone & Duncan 1995); suppressive pool (Reynolds & Heeger 2009; Carandini & Heeger 2012; Heeger 1992)	C; D (softmax Z is this with exponential drive)
Share_j	$P_j^{\gamma} / \text{Budget}_t$	Attention field × drive ÷ suppressive drive (Reynolds & Heeger 2009); attentional weights as ratios (Bundesen 1990)	C; D (with $z_{tj} = \gamma \log P_j + \kappa_t$, softmax $\alpha_{tj} = \text{Share}_j$)
Arousal_t	Vitals-only autonomic index at the readout hour (z(HR)+z(Resp) from decayed input); SOFA_t as sensitivity analysis	Mather & Sutherland 2011; Mather et al. 2016; Aston-Jones & Cohen 2005; Eldar et al. 2013	C that arousal amplifies competition; A for the clinical analogue
γ_t	M0: constant · M1: $\gamma_0 + \gamma_1 \cdot \max_j \log P_j$ · M2: $\gamma_0 + \gamma_1 \cdot \text{Arousal}_t$	Same as above	A
Content_j, Risk_t	v_j; $g(\Sigma\ \text{Share}_j \cdot \text{Content}_j)$ at the t+1 head	Standard attention readout; Kobayashi 2020 (weight × vector)	D

B0a · What γ stands for and the role it plays

Name and meaning. γ is the **sharpness exponent** of the allocation rule: the share an item receives is proportional to its priority *raised to the power γ*, $\text{Share}_j = P_j^{\gamma} / \Sigma_i P_i^{\gamma}$. It is the exponent in the divisive-normalization equation (Heeger, 1992; Reynolds & Heeger, 2009). **[C]** In a Transformer it is the inverse temperature of the softmax: writing $z_{tj} = \gamma \log P_j + \kappa_t$ gives exactly the same weights, so scaling the logits (for example by the length of the query vector q_t) changes γ without changing which hour ranks first. **[D]** Eldar, Cohen & Niv (2013) describe neural gain in the same terms: higher gain increases the contrast between strong and weak inputs, focusing attention; lower gain broadens it. **[C]**

γ	Behaviour	Reading
γ = 1	Share is proportional to priority	Neutral allocation. Nothing selective happens.
γ > 1	Strong items get more than their proportional share, weak items less	Winner-take-more, loser-take-less (the ABC signature)
0 < γ < 1	Shares are flattened toward equal	Broad, unselective attention
γ → ∞	All weight on the single strongest item	Hard winner-take-all

Why γ matters for the project. It is the dial that separates the three rival models. Under M0 γ never moves, so raising one item's priority shrinks every other item by the same factor and nothing selective can appear. Under M1 and M2 γ *moves* with context (the strongest item, or the arousal level), so weak items lose more than strong ones. **Worked numbers:** take priorities 4, 2, 1 and raise the top priority from 4 to 8. With γ fixed, both other hours lose the same log amount (−0.45). When γ rises from 1 to 1.5 the weakest hour loses more (−1.33) than the middle one (−0.98). The slope on log priority equals the change in γ (0.5).

Two cautions about γ

- **γ is not identified from natural data on its own.** From $\log \alpha_j = \gamma \cdot \log P_j + \kappa$ you recover only the product γ·g(severity); a larger γ and a steeper severity-to-priority map look identical. That is why the tests target *changes* in sharpness under an intervention, and why "γ > 1" as an absolute level is not a claim we make.
- **Absolute γ depends on the unit of priority.** Rescaling severity rescales γ. Compare γ across conditions (high versus low arousal, softmax versus sigmoid), never against a universal threshold.

Three rival high-level models, and what each predicts

Because ratios among non-target hours are unchanged by $\text{do}(\text{Priority}_k)$ whenever γ is fixed, selective suppression can only arise if sharpness itself responds to context. That gives three models that make different predictions, so the experiment can decide between them rather than merely "find a pattern". **[D]**

Intervention	M0 · fixed γ (pure normalization)	M1 · saliency gain	M2 · arousal gain (ABC)
$\text{do}(\text{Priority}_k \uparrow)$	Share_k rises; every other hour shrinks by the same factor → slope $\lambda_1 = 0$	Share_k rises; others shrink selectively → $\lambda_1 > 0$ (when P_k becomes the strongest item)	Same as M0 ($\lambda_1 = 0$)
$\text{do}(\text{Arousal}_t \uparrow)$	No change	No change	Sharper allocation: $\lambda_1 > 0$, row entropy falls, order of hours unchanged
Interaction	None	None	Rise in Share_k from $\text{do}(\text{Priority}_k)$ is larger when Arousal_t is high

Slope convention. $r_j = \Delta z_j - \text{mean}_S \Delta z$ (softmax) over $S = \{k < j < t\}$, regressed on $\log P_j$ (or severity_j). Under a sharpness change $\gamma \to \gamma'$, $r_j = (\gamma' - \gamma)(\log P_j - \text{mean})$ so the slope equals $\gamma' - \gamma$. Positive = low-priority hours fall further than high-priority ones. **Placebo set:** for $j < k$ the causal mask forces $\Delta z_j = 0$ exactly, which gives the empirical noise floor. **Layer-1 check:** r_j must be 0 in layer 1 for every $j \neq k$; structure there means a bug. **[D]**

Sigmoid arm. There is no shared normalizer, so no Budget node and no forced sharing; the quantity unavailable under softmax is the total mass $\Sigma_j a_{tj}$. If it stays roughly constant as Priority_k rises, the network has *learned* a capacity budget it was never forced to have, which is the strongest available evidence for the limited-capacity premise. **[D]**

P5 · Localize: counterfactual activation patching with real pairs

What and why

Activation patching runs a model on a base input and a source input and copies an internal activation from one run into the other to see what it carries. Using a *real* second input as the counterfactual goes back to Vig et al. (2020) (causal mediation analysis with real contrastive sentences) and is the standard in circuit work such as Wang et al. (2023). **[C]** Heimersheim & Nanda (2024) is a practical guide to noising versus denoising and to reading the results. **[C]** We drop Meng et al.'s (2022) Gaussian-noise corruption because Zhang & Nanda (2024) found results depend strongly on corruption method and noise level; a noise level only slightly lower than ROME's let the model recover the answer despite the corruption. **[C]** Continuous vitals and labs make noise-scale sensitivity worse, not better. **[A]**

What to patch. S0–S6 are the intervention sites; the table gives the high-level node each one stands for.

Site	What is patched	Tests which edge	Measured	Caution
S0	Input $\hat{x}$ at hour k (splice a donor hour; recompute decay features downstream)	Severity → Priority → everything (C1 manipulation)	Δz_j for $j \neq k$ at readout t	Splice can look like a spike; see plausibility checks
S1	Embedding h_k only (skips decay recomputation)	Same, isolating what hour k contributes as key/value	Δz_j	Report S0 and S1; divergence is itself a result
S2	Residual stream at (L, k), sweep L = 1…N	Where Priority_k is represented	Restoration fraction RE	Exploratory; LOC split only
S3	Head output / MLP output at (L, t)	Attention route versus MLP route	RE	Path patching (Wang et al. 2023)
S4	Add δ to the logit z_{tk} before the normalizer	Share → Risk (does the output depend on allocation)	Change in t+1 logit and in the other α	Logit level only; same in the sigmoid arm
S5	Value vector v_k only	Content versus Priority route (weight versus vector)	RE	Pairs with Kobayashi's weight × norm logic
S6	Residual stream at (L, t), query side (scales q_t)	Arousal_t → γ (sharpness)	Row entropy, slope λ_1	Needs layer ≥ 2 to be non-trivial

Pair construction from your columns

- **Base**: a stay from LOC (localization) or CONF (confirmation); choose $k \geq 3$ and a readout t with $t - k \geq 3$. All rows are pre-onset already (post-onset rows were dropped).
- **Severity class**: severe = SOFA_total ≥ Q90, routine = SOFA_total ≤ Q50, quantiles from the FIT fold (pre-register).
- **Donor hour**: same fold, different subject; match ICULOS bucket (±4 h), measurement-density bucket (±2 features), Unit1/Unit2 pattern, opposite severity class, and the *previous* hour's HR/MAP/Resp/O2Sat within a tolerance so the entry is smooth.
- **Size**: about 2,000 pairs per fold, at most 3 per stay. Verify the noise floor first: two identical runs must give $\Delta z \approx 0$ to float precision (fp32, dropout off).
- **Metric**: RE = (f(patched) − f(base)) / (f(source) − f(base)) at a fixed readout, f = t+1 logit; drop pairs with a tiny denominator.
- **Plausibility**: measure how often raw stays contain isolated one-hour excursions of the spliced size (HR, MAP, Resp); rerun C1 with loose versus context-matched donors and compare slopes.

P6 · Define the high-level model (rival models M0, M1, M2)

The node table and the rival-model table in B0 are the definition. Practical points:

- **Fit on LOC, evaluate on CONF.** The function g (severity → log priority) and each model's sharpness parameters are estimated on LOC from the model's own logits; CONF is untouched until the end. M0 has the fewest parameters; report that.
- **Predictions per pair.** Base priorities P^b and the source value P_k^s: $\text{Share}'_j = P'^{\gamma'}_j / \Sigma_i P'^{\gamma'}_i$, with $\gamma' = \gamma$ under M0 and M2 ($\text{do}(\text{Priority}_k)$ does not move Arousal_t) and the max-dependent γ' under M1. Predicted change vector: Δ

log Share$_j$ for $j \neq k$.

- **Budget is not a stored activation.** It is the softmax denominator, derived from the logits; test it by checking consistency of $\Delta \log Z$, not by patching it.
- **Arousal$_t$ in the network** is the query-side gain at hour t: $z_{tj} = q_t \cdot k_j / \sqrt{d}$, so scaling q_t changes sharpness without changing order. **[D]** The analogy to neural gain is Eldar et al. (2013). **[A]**

P7 · Interchange interventions and IIA

An interchange intervention takes a source run, copies the activation at an aligned site into a base run, and asks whether the output change matches what the high-level model predicts for the corresponding high-level intervention. Interchange intervention accuracy (IIA) is the fraction of pairs where it does (Geiger et al., 2021; 2025). **[C]**

- **IIA$_{share}$ (primary):** Kendall τ between the predicted $\Delta \log$ Share vector and the observed $\Delta \log \alpha$ over $j \in S$; count a match when $\tau \geq$ a pre-registered threshold (default 0.3; a project default, not a literature value).
- **IIA$_{risk}$ (secondary):** agreement in direction (and tertile) of the change in the t+1 logit, using the readout calibrated on LOC.
- **Always report** the random-alignment floor (same dimension), IIA on pairs where the model predicts *no change* separately (a network that ignores interventions scores high there), and results for each of M0, M1, M2.
- **Decision rule:** M_i is supported over M_j only if $IIA_{share}(M_i) - IIA_{share}(M_j)$ exceeds a pre-registered margin with a stay-level bootstrap CI excluding 0, on CONF, and replicates on EXT with the alignment frozen. If none beats the floor, treat the hand-specified alignment as uninterpretable and move to P8; do not call it disconfirming yet.
- **Sigmoid arm:** same procedure; M0/M1/M2 lose their Budget node, so compare on Δz and on $\Sigma_j a_{tj}$.

P8 · DAS (Distributed Alignment Search), explained

The problem DAS solves

A hand-specified alignment says "this variable lives in this head's output" or "in these neurons". That assumes the variable occupies a disjoint, axis-aligned set of units. In a Transformer, features are often spread across directions of the residual stream, so no single neuron or head may carry the variable even if the network represents it. Geiger et al. (2024) motivate DAS exactly this way: brute-force search over neuron sets is intractable and assumes the variable is localized. **[C]**

How it works, step by step

- Pick a site (for Priority$_k$: the residual stream at layer L, hour k; for Arousal$_t$: layer L, hour t). Let $h \in R^d$ be the activation there (d = d_model).
- Learn an **orthogonal matrix R** (a rotation of the basis). In the rotated coordinates Rh, the first m coordinates are declared to be the subspace for the variable; the remaining d − m are the complement.
- **Distributed interchange intervention:** for a base activation h_b and a source activation h_s, build $h^* = R^\top [\, (Rh_s)_{1\ldots m} ; (Rh_b)_{m+1\ldots d} \,]$: swap the subspace, keep the complement, rotate back.
- Continue the forward pass from the site with h^* to get y_{low}. Compute y_{high}: what the high-level model M_i predicts when the variable is set to the source's value.
- **Train R only** (network frozen) by gradient descent so that y_{low} matches y_{high}. R is kept orthogonal by parametrization. Boundless DAS (Wu et al., 2023) also learns the subspace size instead of fixing m.
- **Evaluate IIA on held-out stays.** High IIA says: in some rotated basis, a small subspace carries the variable *and* downstream computation uses it as the high-level model says.

```
for each batch of pairs (base b, source s):
    h_b, h_s = frozen_model.activations_at(L, pos)            # from natural forward passes
    y_high   = M_i.counterfactual(base=b, variable := V(s))   # high-level target
    h_star   = R.T @ concat( (R@h_s)[:m], (R@h_b)[m:] )       # distributed interchange
    y_low    = frozen_model.forward_from(L, pos, h_star)
    loss     = CE_or_MSE(y_low, y_high);  update R only
evaluate IIA on held-out stays; repeat for m in {1, 2, 4, 8, 16}
```

Why DAS can mislead, and the controls that follow

Illusion risk. Makelov et al. (2024) showed that patching a learned subspace can change the output by activating a pathway that is dormant under natural inputs; in their example a one-dimensional subspace patch changed the output while patching the whole component did not. High IIA therefore does not by itself prove the subspace is *used*. **[C]** Wu et al. (2024) published a reply disputing how general the illusion is; the disagreement is unresolved, so we treat it as a risk to control for rather than a settled flaw. **[C]** **Triviality risk.** Sutter et al. (2025): with unconstrained nonlinear alignment maps, any network aligns with any algorithm, so we keep R linear and orthogonal. **[C]**

Control (run every one)	What it rules out
DAS on a randomly initialized network of the same architecture	Alignment found because the map is flexible, not because the network encodes the variable
DAS trained on shuffled counterfactual labels	Fitting noise
Random subspace of the same dimension m, no training	A subspace is not special just because it has m dimensions
Train on LOC, evaluate on CONF (never the same stays); replicate on EXT	Overfitting R to the stays it was trained on
Compare subspace patch with whole-component patch; check that the subspace varies with the variable in *natural* activations	The dormant-pathway illusion

Success rule (pre-register now): DAS counts only if its held-out IIA exceeds all three random controls by the smallest effect of interest. If it does not, treat the ABC alignment as absent and report that as evidence against the hypothesis; without this branch the staged fallback could only ever move toward confirmation. *Refinement:* earlier notes suggested splitting CONF for DAS training; training on LOC and leaving CONF untouched is simpler and stricter. Implementation: the pyvene library supports DAS and Boundless DAS interventions on PyTorch models.

Part C — The weak link: ABC is contested. How the design meets it

The evidence for arousal-biased competition is mixed. Supportive: Mather & Sutherland (2011), Sutherland & Mather (2012), Lee, Itti & Mather (2012). **[C]** Against or qualifying: Keitel et al. (2013) found that competition suppressed attended and unattended stimuli similarly while attention acted as an independent gain, and a Leiden University group reports no arousal-biased competition in focused visuospatial attention. **[C]** This is the project's weakest theoretical link, and a reviewer will say so. The design does not hide it; it is built so that the project's value does not depend on ABC being true.

Principle 1 · ABC is a hypothesis generator, not a prerequisite

The bridge claimed is about a shared computational problem (allocate a limited resource among competing inputs), not about brains and Transformers being the same thing. This is Marr's (1982) computational level of analysis, where the problem and the strategy are described independently of the hardware. **[C]** ABC supplies structured, testable predictions about *how* such an allocation should behave under high arousal; the experiment then checks whether this network behaves that way. A null result says nothing against ABC in humans, only that this system does not implement amplified competition.

Principle 2 · A ladder of claims, so that each rung is reportable on its own

Rung	Claim	Depends on ABC?	Decided by
0	Attention is severity-sensitive after controls	No	P1–P4 (correlation, controls)
1	The network implements allocation by normalization (M0 beats the floor)	No (this is the Keitel-type "gain + non-selective competition" pattern)	P7 IIA for M0; sigmoid contrast (Part D)
2	Sharpness depends on context (M1)	Partly (saliency-dependent competition: Itti & Koch 2000)	do(Priority$_k$) slope $\lambda_1 > 0$ beyond SESOI; IIA M1 > M0
3	Sharpness is gated by arousal (M2)	**Yes: the ABC-specific claim**	do(Arousal$_t$) slope, severity × arousal interaction, IIA M2 > M0 and M1 by the pre-registered margin

If rung 3 fails, rungs 0–2 still stand and still answer the original question about whether attention heatmaps can be trusted in this clinical model. The paper is then a rigorous test of a hypothesis with a clear outcome, which is a legitimate contribution (Platt, 1964, on rival hypotheses and strong inference). **[C]**

Principle 3 · Meet each specific weakness with a specific safeguard

Weak link	Why it is a problem	Safeguard in the design
Empirical support is mixed	A positive result could be an artefact; a null could be dismissed as a bad analogy	Rival models M0/M1/M2 with M0 as the null; report all three IIA values; claim M2 only if it beats both by a margin on CONF and replicates on EXT
Gain and competition are separable (Keitel et al. 2013)	"Winner gets more" can arise from gain alone, with no selective suppression	M0 *is* that gain-plus-uniform-suppression pattern; selectivity must beat it (slope $\lambda_1 > 0$ against the exact null $\lambda_1 = 0$)
Interaction effects are hard to interpret	A winner-take-more pattern can arise from the scale of the measure rather than from the mechanism	Work on the logit scale, where renormalization is additive; use exact nulls (common shift, placebo set j < k, layer-1 zero) instead of interaction terms on bounded outcomes
Severity = priority is an analogy	Hour-level SOFA may not be what the network treats as salient	Estimate g(severity → log priority) from the network itself on LOC; test several definitions of severity (SOFA_total, component scores, deviation from baseline) as pre-registered sensitivity analyses with FDR control
Acuity = arousal is an analogy	Any variable correlated with acuity could produce the interaction	Primary index uses vitals only (not SOFA); run (i) a **placebo arousal**: the same index circularly shifted within stay, which must not gate sharpness; (ii) a **dose-response** check across arousal quantiles; (iii) SOFA$_t$ as an alternative index

Weak link	Why it is a problem	Safeguard in the design
Researcher degrees of freedom	Many thresholds, sites and definitions invite post-hoc choices	Pre-register the arousal index, the severity classes, τ threshold, margins and k heads before touching CONF (Nosek et al., 2018). **[C]**
Model may implement ABC-like structure by accident of training	Matching a prediction is not proof of mechanism	Random-network and shuffled-label controls for IIA and DAS; sigmoid contrast to see whether the pattern needs the zero-sum constraint

What would count as evidence against ABC-style amplification here (decide now)

- M2 does not exceed M0 and M1 on IIA_{share} by the pre-registered margin on CONF, **and** the do($Arousal_t$) slope is equivalent to zero (TOST against the smallest effect of interest) with a noise floor well below that bound.
- Placebo arousal gates sharpness as strongly as real arousal.
- The effect appears in the untrained-network control.
- State the conclusion at the right scope: "this system does not show arousal-gated competition", not "ABC is false".

Part D — Sigmoid arm: analysis plan

Why this arm exists. Under softmax, "other hours fall when one rises" is partly arithmetic. In the sigmoid arm each key is weighted by its own logit and nothing is forced to sum to 1, so any competition has to be *learned*. The pair isolates the normalizer. Ramapuram et al. (2025) analyse sigmoid self-attention theoretically and report that it retains the universal-approximation property established for softmax Transformers by Yun et al. (2020), so a difference between arms cannot be put down to lost representational capacity. **[C]**

The two arms side by side

	Softmax arm	Sigmoid arm
Weight	$\alpha_{tj} = \exp(z_{tj}) / \Sigma_{i \le t} \exp(z_{ti})$	$a_{tj} = \sigma(z_{tj} - \log T_t)$, T_t = number of visible keys
Constraint	$\Sigma_j \alpha_{tj} = 1$ (shared budget, forced)	$\Sigma_j a_{tj}$ free (budget, if any, must be learned)
Role of log T	—	Keeps output scale matched to softmax at initialization ($a \approx 1/T$); a deliberate t-dependent term, so keep a query-position covariate
Screening DV (P1)	clr(α)	log a_{tj}, uncentered
Causal DV (P5–P7)	Δz, centered on the common shift c	Δz, **not** centered
Exact null under do($Priority_k$)	$\Delta z_j = c$ for all $j \neq k$ (common shift); $r_j = 0$	$\Delta z_j = 0$ for all $j \neq k$ (no shift at all)
High-level model	Budget + Share (see B0)	No Budget node: independent gates, plus any learned inhibition

What to measure in the sigmoid arm

- **Common shift.** mean of Δz_j over $j \in S$ after do($Priority_k \uparrow$). Zero means independent gates. A negative value cannot be arithmetic, so it is learned lateral inhibition. Compare its size with the forced shift c in the softmax arm: similar sizes mean the competition is learned, not imposed. **[D]**
- **Selectivity slope.** λ_1 of Δz_j on log $priority_j$, same convention as softmax (positive = low-priority hours suppressed more). **[D]**
- **Total mass.** $M_t = \Sigma_{j \le t} a_{tj}$. Regress M_t on window priority composition, T_t and measurement density. A coefficient near zero on composition means the network keeps total attention roughly constant: a budget it was never forced to have, which is the strongest available evidence for the limited-capacity premise and is impossible to obtain under softmax. **[D]**

- **Analyse logits, not weights.** Near saturation $\sigma'(x) = a(1 - a)$ is small, so changes in a understate changes in z; Δz is the quantity that is the same in both arms.
- **Same interventions.** S0–S6 apply unchanged; S4 (adding δ to a logit) needs no renormalization step. Layer-1 check: Δz_j must be exactly 0 for $j \neq k$.

Idealized expectations. Softmax shows a forced common shift; a slope above zero adds selectivity. Sigmoid with independent gates shows nothing; anything else is learned.

Four-outcome interpretation (softmax × sigmoid)

"Pattern present" means a selectivity slope (or, for sigmoid, a shift or slope) beyond the smallest effect of interest and above the control band, under the intervention the supported model predicts.

Softmax	Sigmoid	Reading	Check before believing it
Present	Present	Learned, normalizer-independent competition. Strongest support; with constant total mass it is a learned capacity budget.	Both arms pass the AUPRC gate; placebo and untrained controls clean in both
Present	Absent	Competition is scaffolded by the zero-sum constraint: selective allocation appears only when allocation is forced. Real and interesting; weakens the "shared computational problem" framing, because the mechanism depends on the normalizer.	Sigmoid arm is not dead: total mass not collapsed, AUPRC gate passed, scale not drifting
Absent	Present	Most likely a floor effect in softmax: if weak hours are driven to $\alpha \approx 0$, log α has no room to fall further and the slope is attenuated. A genuinely different (gating) solution is the alternative.	Fraction of $\alpha < 10^{-6}$ in S; re-analyse softmax on Δz (no floor); inspect total-mass regression in the sigmoid arm
Absent	Absent	No learned selective competition: attention is severity-sensitive but proportional (M0). A real null against amplified competition in this system.	Equivalence test (TOST) against the smallest effect of interest, and a noise floor clearly below it; otherwise report "underpowered"

Matching and training checklist (so the contrast is fair)

- Identical embedding module, initialization seed, optimizer, schedule, batch size, loss, positional encoding and mask; only the normalizer (and the $-\log T$ bias) differ. Assert parameter-tensor equality at step 0 for everything else.
- Do not renormalize sigmoid weights afterwards; do not add sparsity or entropy penalties to either arm; if QK-norm is used, use it in both.
- Longer warmup for the sigmoid arm (about 2×); log mean total mass every epoch. If it collapses toward 0 attention is being bypassed; if it grows without bound, scale is drifting (clip gradients).
- If the sigmoid arm needs a different learning rate to clear the performance gate, tune and report it as a limitation. A sigmoid arm that fails the gate cannot serve as a null.
- Rollout is not defined for the sigmoid arm (rows do not sum to 1), so interpret it only as raw linear influence if at all; the primary analyses do not use rollout.

What to change on slides 7 and 8

Your current slides are close; four lines need updating to match this design.

Slide / point	Current text	Suggested text
7 · 1	Per-head, centered-log-ratio-transformed attention weights … not raw rollout	Per-head attention, log-ratio (clr) transformed for screening; pre-normalizer logits Δz for the causal test
7 · 3	Mixed-effects regression, patient as a random effect	Two-stage regression: per-stay fit, then random-effects meta-analysis; heads chosen on a discovery split
7 · 4	Untrained-network, shuffled-label, circular-shift baselines	Add: AUPRC gate first; controls define the noise floor an effect must exceed

Slide / point	Current text	Suggested text
8 · 2	A causal graph: Severity → Priority → shared Budget → Share → Risk	Three rival high-level models (M0 normalization, M1 saliency gain, M2 arousal gain), each grounded in the normalization model of attention and ABC
8 · 3	Swap a severe hour's activation into a routine sequence …	Interchange interventions; ask which model's predictions the network matches, on held-out CONF stays
8 · 4	If a hand-specified alignment is ambiguous, DAS learns the subspace	DAS: learned orthogonal rotation; accepted only if it beats random-network, shuffled-label and random-subspace controls

Limits, and what is not established

- Severity → priority and acuity → arousal are analogies; ABC was developed for perception and memory, not ICU time series. **[A]**
- ABC's empirical support is mixed (Keitel et al. 2013; a Leiden null report). The experiment can legitimately come back null; Part C sets out how the design handles that and what each outcome means.
- "Softmax = divisive normalization with exponential drive" is our own algebra **[D]**; the neuroscience literature establishes divisive normalization in general, not this identity for Transformers.
- Thresholds (Q90/Q50, $\tau \geq 0.3$, margins, k heads) are pre-registration defaults, not literature values; set them before looking at CONF.
- One reference is a partial citation (marked ‡ below); confirm authors and venue before it goes into a paper or slide.

References

Abnar, S., & Zuidema, W. (2020). Quantifying attention flow in transformers. *ACL 2020*.

Adebayo, J., Gilmer, J., Muelly, M., Goodfellow, I., Hardt, M., & Kim, B. (2018). Sanity checks for saliency maps. *NeurIPS 31*.

Aitchison, J. (1982). The statistical analysis of compositional data. *Journal of the Royal Statistical Society B, 44*(2), 139–177.

Aston-Jones, G., & Cohen, J. D. (2005). An integrative theory of locus coeruleus–norepinephrine function: adaptive gain and optimal performance. *Annual Review of Neuroscience, 28*, 403–450.

Bates, D., Mächler, M., Bolker, B., & Walker, S. (2015). Fitting linear mixed-effects models using lme4. *Journal of Statistical Software, 67*(1), 1–48.

Beckers, S., & Halpern, J. Y. (2019). Abstracting causal models. *AAAI 2019*.

Bundesen, C. (1990). A theory of visual attention. *Psychological Review, 97*(4), 523–547.

Burke, D. L., Ensor, J., & Riley, R. D. (2017). Meta-analysis using individual participant data: one-stage and two-stage approaches, and why they may differ. *Statistics in Medicine, 36*(5), 855–875.

Carandini, M., & Heeger, D. J. (2012). Normalization as a canonical neural computation. *Nature Reviews Neuroscience, 13*(1), 51–62.

Desimone, R., & Duncan, J. (1995). Neural mechanisms of selective visual attention. *Annual Review of Neuroscience, 18*, 193–222.

DerSimonian, R., & Laird, N. (1986). Meta-analysis in clinical trials. *Controlled Clinical Trials, 7*(3), 177–188.

Eldar, E., Cohen, J. D., & Niv, Y. (2013). The effects of neural gain on attention and learning. *Nature Neuroscience, 16*(8), 1146–1153.

Geiger, A., Lu, H., Icard, T., & Potts, C. (2021). Causal abstractions of neural networks. *NeurIPS 34*.

Geiger, A., Wu, Z., Potts, C., Icard, T., & Goodman, N. (2024). Finding alignments between interpretable causal variables and distributed neural representations. *CLeaR 2024, PMLR 236*, 160–187.

Geiger, A., Ibeling, D., Zur, A., Chaudhary, M., Chauhan, S., Huang, J., Arora, A., Wu, Z., Goodman, N., Potts, C., & Icard, T. (2025). Causal abstraction: a theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research, 26*(83).

Gelman, A., & Hill, J. (2007). *Data Analysis Using Regression and Multilevel/Hierarchical Models*. Cambridge University Press.

Haviv, A., Ram, O., Press, O., Izsak, P., & Levy, O. (2022). Transformer language models without positional encodings still learn positional information. *Findings of EMNLP 2022*.

Heeger, D. J. (1992). Normalization of cell responses in cat striate cortex. *Visual Neuroscience, 9*(2), 181–197.

Heimersheim, S., & Nanda, N. (2024). How to use and interpret activation patching. *arXiv:2404.15255*.

Hewitt, J., & Liang, P. (2019). Designing and interpreting probes with control tasks. *EMNLP-IJCNLP 2019*.

Itti, L., & Koch, C. (2000). A saliency-based search mechanism for overt and covert shifts of visual attention. *Vision Research, 40*(10–12), 1489–1506.

Jain, S., & Wallace, B. C. (2019). Attention is not explanation. *NAACL-HLT 2019*.

Keitel, C., Andersen, S. K., Quigley, C., & Müller, M. M. (2013). Independent effects of attentional gain control and competitive interactions on visual stimulus processing. *Cerebral Cortex, 23*(4), 940–946.

Kobayashi, G., Kuribayashi, T., Yokoi, S., & Inui, K. (2020). Attention is not only a weight: analyzing transformers with vector norms. *EMNLP 2020*.

Kobayashi, G., Kuribayashi, T., Yokoi, S., & Inui, K. (2021). Incorporating residual and normalization layers into analysis of masked language models. *EMNLP 2021*.

Lakens, D. (2017). Equivalence tests: a practical primer for t tests, correlations, and meta-analyses. *Social Psychological and Personality Science, 8*(4), 355–362.

Lee, T.-H., Itti, L., & Mather, M. (2012). Evidence for arousal-biased competition in perceptual learning. *Frontiers in Psychology, 3*, 241.

Leiden University group (S. Nieuwenhuis, senior author). *No arousal-biased competition in focused visuospatial attention*. ‡ Partial citation from a university repository record: confirm authors and venue before citing.

Makelov, A., Lange, G., Geiger, A., & Nanda, N. (2024). Is this the subspace you are looking for? An interpretability illusion for subspace activation patching. *ICLR 2024*.

Mather, M., & Sutherland, M. R. (2011). Arousal-biased competition in perception and memory. *Perspectives on Psychological Science, 6*(2), 114–133.

Mather, M., Clewett, D., Sakaki, M., & Harley, C. W. (2016). Norepinephrine ignites local hotspots of neuronal excitation: how arousal amplifies selectivity in perception and memory. *Behavioral and Brain Sciences, 39*, e200.

Marr, D. (1982). *Vision: A Computational Investigation into the Human Representation and Processing of Visual Information*. W. H. Freeman.

Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). The preregistration revolution. *Proceedings of the National Academy of Sciences, 115*(11), 2600–2605.

Platt, J. R. (1964). Strong inference. *Science, 146*(3642), 347–353.

Ramapuram, J., Danieli, F., Dhekane, E., Weers, F., Busbridge, D., Ablin, P., Likhomanenko, T., et al. (2025). Theory, analysis, and best practices for sigmoid self-attention. *ICLR 2025*.

Yun, C., Bhojanapalli, S., Rawat, A. S., Reddi, S., & Kumar, S. (2020). Are transformers universal approximators of sequence-to-sequence functions? *ICLR 2020*.

Meng, K., Bau, D., Andonian, A., & Belinkov, Y. (2022). Locating and editing factual associations in GPT. *NeurIPS 35*.

Michel, P., Levy, O., & Neubig, G. (2019). Are sixteen heads really better than one? *NeurIPS 32*.

Miller, E. K., & Cohen, J. D. (2001). An integrative theory of prefrontal cortex function. *Annual Review of Neuroscience, 24*, 167–202.

Reynolds, J. H., & Heeger, D. J. (2009). The normalization model of attention. *Neuron, 61*(2), 168–185.

Singer, M., Deutschman, C. S., Seymour, C. W., et al. (2016). The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA, 315*(8), 801–810.

Sutherland, M. R., & Mather, M. (2012). Negative arousal amplifies the effects of saliency in short-term memory. *Emotion, 12*(6), 1367–1372.

Sutter, D., Minder, J., Hofmann, T., & Pimentel, T. (2025). The non-linear representation dilemma: is causal abstraction enough for mechanistic interpretability? *NeurIPS 2025*.

Theiler, J., Eubank, S., Longtin, A., Galdrikian, B., & Farmer, J. D. (1992). Testing for nonlinearity in time series: the method of surrogate data. *Physica D, 58*(1–4), 77–94.

Vig, J., Gehrmann, S., Belinkov, Y., Qian, S., Nevo, D., Singer, Y., & Shieber, S. (2020). Investigating gender bias in language models using causal mediation analysis. *NeurIPS 33*.

Vincent, J.-L., de Mendonça, A., Cantraine, F., et al. (1998). Use of the SOFA score to assess the incidence of organ dysfunction/failure in intensive care units. *Critical Care Medicine, 26*(11), 1793–1800.

Voita, E., Talbot, D., Moiseev, F., Sennrich, R., & Titov, I. (2019). Analyzing multi-head self-attention: specialized heads do the heavy lifting, the rest can be pruned. *ACL 2019*.

Wang, K. R., Variengien, A., Conmy, A., Shlegeris, B., & Steinhardt, J. (2023). Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. *ICLR 2023*.

Wiegreffe, S., & Pinter, Y. (2019). Attention is not not explanation. *EMNLP-IJCNLP 2019*.

Wu, Z., Geiger, A., Icard, T., Potts, C., & Goodman, N. D. (2023). Interpretability at scale: identifying causal mechanisms in Alpaca. *NeurIPS 36*.

Wu, Z., Geiger, A., Huang, J., Arora, A., Icard, T., Potts, C., & Goodman, N. D. (2024). A reply to Makelov et al. (2023)’s “interpretability illusion” arguments. *arXiv:2401.12631*.

Zhang, F., & Nanda, N. (2024). Towards best practices of activation patching in language models: metrics and methods. *ICLR 2024*.